Mathematical Logic

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November 8, 2026

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1 Mathematical Logic

I Proof Theory

i Deductive system, proof system, formal system, formal proof system, proof calculus, or deductive calculus

A deductive system, proof system, formal system, or formal proof system Γ over a formal language $\mathcal{L}$ consists of axioms and rules of inference, where:

- An axiom is a formula in $\mathcal{L}$ assumed to be derived in Γ.
- An axiom schema is an expression with schema variables, each designated with a type of symbols in $\mathcal{L}$, that the expression obtained by replacing each schema variable with an arbitrary symbol of its type, called an application of the axiom schema, is assumed to be an axiom.
- A rule of inference in Hilbert style is in the form $\frac{P_1, P_2, \dots, P_n}{Q}$ where $P_1, P_2, \dots, P_n, Q$ are expressions sharing the same set of metaformulas such that any expressions obtained by replacing each metaformula with an arbitrary formula in $\mathcal{L}$ are formulas in $\mathcal{L}$ and the expressions obtained by such replacement of $P_1, P_2, \dots, P_n$ are assumed to derive the expression obtained by such replacement of Q, in which $P_1, P_2, \dots, P_n$ and their replacements are called preimises and Q and their replacements called conclusion. Other styles exist and have the same expressive power as Hilbert style for propositional logic and predicate logic.

ii (Formal) derivation or proof

Given proof system Γ over language $\mathcal{L}$ and formulas $S_1, S_2, \dots S_n, T \in \mathcal{L}$, a (formal) derivation or proof from $S_1, S_2, \dots S_n$ to T is a sequence of formulas $\langle F_i \rangle_{i=1}^{m}$ such that each F_i is either

- an axiom of Γ,
- an element of $\{S_1, S_2, \dots S_n\}$, or
- can be derived from some of the earlier formulas in the sequence using a rule of inference,

and that $F_m = T$.

$S_1, S_2, \dots S_n$ are called premises, T is called conclusion, and this process is called argumentation.

That there exists a derivation from $S_1, S_2, \dots S_n$ to T is denoted as $S_1, S_2, \dots S_n \vdash T$ and called $S_1, S_2, \dots S_n$ derive or prove T.

Thus, for any axiom α, $\vdash \alpha$.

Any formula α such that $\vdash \alpha$ is called derived or a theorem and is also denoted as $\Gamma \vdash \alpha$.

iii Finitarity

A rule of inference is finitary iff its premises are finite.

The rules of inference and sequences in formal proofs are required to be finite for finitary deductive systems.

iv Soundness

A deductive system is sound if every formula that can be derived from the system is a tautology in the language.

v Syntactic consequence

For formulas $S_1, S_2, \dots S_n, T$ in a language $\mathcal{L}$, T is a syntactic consequence of $S_1, S_2, \dots S_n$, denoted as $S_1, S_2, \dots S_n \vdash T$, iff there exists a sound proof system over $\mathcal{L}$ in which $S_1, S_2, \dots S_n \vdash T$.

vi Independent formula

A formula P such that neither $\vdash P$ nor $\vdash \neg P$.

vii (Syntactical) completeness

A deductive system is (syntactically) complete if for every formula in the language, either it or its negation can be derived from the system.

viii Semantical completeness

A deductive system is semantically complete if every tautology in the language can be derived from the system.

ix Interpretability

If a deductive system S over language L and a deductive system T over language M are such that there exists a function $H : L \to M$ such that $S \vdash l \implies T \vdash H(l)$, then we say T inteprets S, S is weaker than T, or T is stronger than S.

x Computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable deductive system

A deductive system is computably enumerable (c.e.), recursively enumerable (r.e.), semidecidable, partially decidable, listable, provable or Turing-recognizable iff the set of all axioms of it is computably enumerable.

If a deductive system is computably enumerable and its rules of inferences are finitary, it is finitary.

II Model Theory

i Signature

A signature $\sigma = (\mathcal{P}, \mathcal{F})$ is a pair of indexed sets,

- a set of sets of predicate symbols (also called relation symbols) $\mathcal{P} = \{P_n \mid n \in \mathbb{N}_0\}$ with P_n being a possibly infinite set of predicate symbols of arity n,
- a set of sets of function symbols $\mathcal{F} = \{F_n \mid n \in \mathbb{N}_0\}$ with F_n being a possibly infinite set of function symbols of arity n, with any element of F_0 called a constant.

ii (Logical or model-theoric) structure

A (logical or model-theoric) structure can be defined with a triple (A, σ, I) of a set called domain, universe, carrier (set), underlying set, domain of discourse, or universe of discourse, A, denoted as $|\mathcal{A}|$ or simply $\mathcal{A}$, a signature σ, and an interpretation function I that, maps for all $n \in \mathbb{N}_0$, maps each $f \in F_n$ to an n-ary operation $I(f)$ on A, called interpretation of f and often denoted as f, and for all $n \in \mathbb{N}_0$, maps each $p \in P_n$ to a subset $I(p)$ of A^n, called interpretation of p and often denoted as p, in which $A^0 = \{\varnothing\}$.

A structure with signature σ is called a σ-structure. A structure with signature σ of a language L is called a L-structure.

iii Substructure

A structure $\mathcal{B} = (B, \sigma_B, I_B)$ is a substructure of a structure $\mathcal{A} = (A, \sigma_A, I_A)$, that is, $\mathcal{A}$ is an extension of $\mathcal{B}$, denoted as $\mathcal{B} \subseteq \mathcal{A}$, iff

- $B \subseteq A$,
- $\sigma_B = \sigma_A$,
- for every $n \in \mathbb{N}_0$, for every function symbol $f \in F_n$, $I_B(f) = I_A(f) \restriction B^n$, and
- for every $n \in \mathbb{N}_0$, for every relation symbol $p \in P_n$, $I_B(p) = I_A(p) \cap B^n$.

iv Elementary substructure

A structure $\mathcal{B} = (B, \sigma_B, I_B)$ is an elementary substructure of a structure $\mathcal{A} = (A, \sigma_A, I_A)$, that is, $\mathcal{A}$ is an elementary extension of $\mathcal{B}$, denoted as $\mathcal{B} \prec \mathcal{A}$, iff, with the language of σ_B denoted as $\mathcal{L}$,

- $\mathcal{B} \subseteq \mathcal{A}$,
- for every n-ary formula $\varphi \in \mathcal{L}$, for every $a \in B^n$, $\mathcal{B} \vdash \varphi[a] \iff \mathcal{A} \vdash \varphi[b]$.

v Closed subset

For a structure $\mathcal{A} = (A, \sigma, I)$, a subset $B \subseteq A$ is called closed iff, for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $b \in B^n$, $I(f)(b) \in B$.

For any subset B of A, there exists a smallest closed subset that is a superset of B, called the hull of B or the closed subset derived by B, denoted as $\langle B \rangle_{\mathcal{A}}$.

vi Cardinality

The cardinality of a structure is defined as the cardinality of its domain.

vii Homomorphism

A ($\sigma-$)homomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B)$ is a map $h : A \to B$ such that,

- for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$h(I_A(f)(a_1, a_2, \ldots, a_n)) = I_B(f)(h(a_1), h(a_2), \ldots, h(a_n)),$$

- for all $n \in \mathbb{N}_0$, for any $p \in P_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$(a_1, a_2, \ldots, a_n) \in I_A(p) \implies (h(a_1), h(a_2), \ldots, h(a_n)) \in I_B(p).$$

For any $a \in A^n$ with $a = (a_1, a_2, \ldots, a_n)$, $(h(a_1), h(a_2), \ldots, h(a_n))$ is denoted as $h(a)$.

viii Strong homomorphism

A strong ($\sigma-$)homomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a map $h : A \to B$ such that,

- for all $n \in \mathbb{N}_0$, for any $f \in F_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$h(I_A(f)(a_1, a_2, \ldots, a_n)) = I_B(f)(h(a_1), h(a_2), \ldots, h(a_n)),$$

- for all $n \in \mathbb{N}_0$, for any $p \in P_n$ and $a_1, a_2, \ldots, a_n \in A^n$,

$$(a_1, a_2, \ldots, a_n) \in I_A(p) \iff (h(a_1), h(a_2), \ldots, h(a_n)) \in I_B(p).$$

ix Embedding

A ($\sigma-$)embedding from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a strong homomorphisms h from $\mathcal{A}$ to $\mathcal{B}$ that is injective.

x Isomorphism

A ($\sigma-$)isomorphism from a structure $\mathcal{A} = (A, \sigma, I_A)$ to a structure $\mathcal{B} = (B, \sigma, I_B$ is a strong homomorphisms h from $\mathcal{A}$ to $\mathcal{B}$ that is bijective.

xi Isomorphism-invariance

An n-free variable formula φ is isomorphism-invariant between two models $\mathcal{M}, \mathcal{N}$ with domains M, N iff for every isomorphism h between $\mathcal{M}, \mathcal{N}$, for any $a \in M^n$:

$$\mathcal{M} \vdash \varphi[a] \iff \mathcal{N} \vdash \varphi[h(a)].$$

xii Definability

Let L be a first-order language, $\mathcal{M}$ be a L-structure with domain M, X be a fixed subset of M, and m be a positive integer.

- A set $A \subseteq M^m$ is definable in $\mathcal{M}$ with parameters from X iff there exists a formula φ of $m + n$ free variables and $b_1, b_2, \ldots, b_n \in X$ such that $\forall a_1 \in M \forall a_2 \in M \ldots \forall a_m \in M$, $(a_1, a_2 \ldots, a_m) \in A$ iff $\mathcal{M} \vDash \varphi[a_1, a_2, \ldots, a_m, b_1, b_2, \ldots, b_n]$, and $\varphi[a_1, a_2, \ldots, a_m, b_1, b_2, \ldots, b_n]$ with parameters $b_1, b_2, \ldots, b_n$ is called defining formula.
- A set is definable in $\mathcal{M}$ without parameter iff it is definable in $\mathcal{M}$ with parameters from $\varnothing$.
- A function is definable in $\mathcal{M}$ (with some parameters) iff its graph is definable in $\mathcal{M}$ (with those parameters).
- An element $a \in M^m$ is definable in $\mathcal{M}$ (with some parameters) iff $\{a\}$ is definable in $\mathcal{M}$ (with those parameters).
- Set of definable sets: Def($\mathcal{M}$) is the union for all $m \in \mathbb{N}$ of the set of subsets of M^m that is definable in $\mathcal{M}$ (with any parameter or without parameter).

III Propositional logic (PL), statement logic, sentential logic, propositional calculus, statement calculus, sentential calculus, or zeroth-order logic

i Syntax

Given

- a set of connectives, logical connectives, logical operators, truth-functional connectives, truth-functors, or propositional connectives $\mathcal{O} = \{\neg, \wedge, \vee\}$ with unary negation $\neg$, binary conjunction $\wedge$, and binary disjunction $\vee$, (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly), and
- a set of atomic propositions, atomic formulas, atomic sentences, atoms, proposition letters, sentence letters, propositional variables, or variables $\mathcal{P}$ with $\mathcal{P} \cap \mathcal{O} = \varnothing$ and $\mathcal{O} \cup \mathcal{P}$ be the alphabet of $\mathcal{L}$,

the formal language $\mathcal{L}$ of propositional logic is defined by:

- $\mathcal{P} \subseteq \mathcal{L}$,
- $\neg P \in \mathcal{L}$ if $P \in \mathcal{L}$,
- $(P \wedge Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \vee Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$, and
- nothing else is in $\mathcal{L}$.

Any $w \in \mathcal{L}$ is called a well-formed formula (WFF or wff), or simply formula.

ii Semantics

For a given propositional logic, its interpretation (or valuation) is defined as an interpretation (or valuation) function of its set of atomic propositions.

For a given set of atomic propositions $\mathcal{P}$, an interpretation (or valuation) function $\mathcal{I}$ maps every formula in $\mathcal{P}$ to a semantic value in $\{1, 0\}$ (with 1 meaning true and 0 meaning false), denoted, for each $P \in \mathcal{P}$, as

$$\mathcal{I}(P) = B.$$

For a given propositional logic with formal language $\mathcal{L}$ and interpretation $\mathcal{I}$, the interpretation function $\mathcal{J}$ over $\mathcal{L}$ is recursively defined as

- for any $P \in \mathcal{P}$, $\mathcal{J}(P) = \mathcal{I}(P)$,
- for any $P \in \mathcal{L}$, $\mathcal{J}(\neg P) = \neg \mathcal{J}(P)$,
- for any $P, Q \in \mathcal{L}$, $\mathcal{J}(P \wedge Q) = \mathcal{J}(P) \wedge \mathcal{J}(Q)$, and
- for any $P, Q \in \mathcal{L}$, $\mathcal{J}(P \vee Q) = \mathcal{J}(P) \vee \mathcal{J}(Q)$.

For any $P \in \mathcal{L}$, we write $\mathcal{I} \vDash P$ if $\mathcal{J}(P) = 1$ and $\mathcal{I} \nvDash P$ if $\mathcal{J}(P) = 0$.

Alternatively, with structure, as expressed in the language of first-order logic, we can define each atomic formula in propositional logic as a nullary predicate symbol, that there is no function symbol, and that the interpretation structure maps each nullary

predicate symbol to a subset of $\{\varnothing\}$, that is, either $\{\varnothing\}$ or $\varnothing$, and with $\varnothing \in \{\varnothing\}$ and $\varnothing \notin \varnothing$, an atomic formula mapped to $\{\varnothing\}$ is true and an atomic formula mapped to $\varnothing$ is false.

iii Semantic consequence

Given formulas $S_1, S_2, \dots S_n, T$, $S_1, S_2, \dots S_n \vDash T$, called that T is a semantic consequence of $S_1, S_2, \dots S_n$, means that for any interpretation function $\mathcal{I}$, if $\mathcal{I} \vDash S_1, \mathcal{I} \vDash S_2, \dots, \mathcal{I} \vDash S_n$, then $\mathcal{I} \vDash T$.

iv Equivalence of syntactic and semantic consequences

For any formulas $S_1, S_2, \dots S_n, T$,

$$S_1, S_2, \dots S_n \vdash T \iff S_1, S_2, \dots S_n \vDash T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

v Classification of formulas

A formula P is called (logically) valid or a tautology, denoted as $\top$, iff for any interpretation function $\mathcal{I}$, $\mathcal{I} \vDash P$.

A formula P is called (logically) invalid or a contradiction, denoted as $\bot$, iff for any interpretation function $\mathcal{I}$, $\mathcal{I} \nvDash P$.

A formula P is called contingent iff it is neither a tautology nor a contradiction.

A formula P is called satisfiable iff it is not a contradiction.

vi Typical rules of inference

For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator.

IV First-order logic (FOL), predicate logic, predicate calculus, or quantificational logic

Proportional logic and predicate logic are collectively called classical logic.

i Syntax

Given logic symbols,

- (either one of) the quantifier symbols $\forall$ for universal quantification and $\exists$ for existential quantification,

- a set of logical connectives $\mathcal{O} = \{\neg, \wedge, \vee, \implies\}$ with unary negation $\neg$, binary conjunction $\wedge$, binary disjunction $\vee$, and binary material implication $\implies$, (or any functionally complete set of logical operators in Boolean algebra with related definitions changed accordingly),
- a set of punctuation symbols $\mathcal{N}$ containing $(,),$, and maybe other symbols, which do not carry meaning themselves but to help parse and read the language,
- an infinite set of variables $\mathcal{V}$,
- an equality symbol = for first-order logic with equality,

and a signature $\sigma = (\mathcal{P}, \mathcal{F})$ with the equality symbol $= \in P_2$ for first-order logic with equality, the set of terms $\mathcal{T}$ is defined by:

- $\mathcal{V} \subseteq \mathcal{T}$, and
- for all $n \in \mathbb{N}_0$, for all $t \in \mathcal{T}^n$, for all $f \in F_n$, $f(t) \in \mathcal{T}$,

the set of atomic formulas $\mathcal{C}$ is defined as

$$\mathcal{C} = \bigcup_{n \in \mathbb{N}_0} \{p(t) \mid t \in \mathcal{T}^n \wedge p \in P_n\},$$

and the formal language $\mathcal{L}$ of first-order logic is defined by:

- $\mathcal{C} \subseteq \mathcal{L}$
- $\neg P \in \mathcal{L}$ if $P \in \mathcal{L}$,
- $(P \wedge Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \vee Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$,
- $(P \implies Q) \in \mathcal{L}$ if $P, Q \in \mathcal{L}$, and
- $\forall x P, \exists x P \in \mathcal{L}$ if $P \in \mathcal{L}$ and $x \in \mathcal{V}$,
- nothing else is in $\mathcal{L}$.

Any $w \in \mathcal{L}$ is called a well-formed formula (WFF or wff) or simply formula.

A variable in a formula is either free or bounded, which are defined as follows for formulas P and Q:

- A variable in an atomic formula is free.
- A variable is free in $\neg P$ if it is free in formula P.
- A variable is free in $P \wedge Q$ if it is free in formula P or Q.
- A variable is free in $P \vee Q$ if it is free in formula P or Q.
- A variable is free in $P \implies Q$ if it is free in formula P or Q.

- A variable x is free in $\forall y P$ and $\exists y P$ if x is not the same symbol as y and x is free in formula P.
- No other variable in a formula is free.

A formula with no free variable is called a sentence or a closed formula.

A formula P with free variables $x_1, x_2, \ldots, x_n$ is denoted as $P[x_1, x_2, \ldots, x_n]$ or $P(x_1, x_2, \ldots, x_n)$.

The cardinalities of σ and $\mathcal{L}$ are both defined as the cardinality of $\left(\bigcup_{u \in \mathcal{P}} u\right) \cup \left(\bigcup_{u \in \mathcal{F}} u\right)$.

ii Semantics

For a given first-order logic, its interpretation (or valuation) is defined as $\mathcal{I} = \mathcal{A}, s$ with an interpretation (or valuation) structure $\mathcal{A} = (A, \sigma, I)$ of the same signature as the first-order logic with the constraint for first-order logic with equality that

$$I(=) = \{(a, a) \mid a \in A\}$$

and a variable assignment function s from the variables $\mathcal{V}$ to A.

For any formula P, $\mathcal{A}, s \vDash P$ denotes that P is true in $\mathcal{A}$ under assignment s, and $\mathcal{A}, s \nvDash P$, denotes that P is false in $\mathcal{A}$ under assignment s.

For a given first-order logic with formal language $\mathcal{L}$ and interpretation $\mathcal{A}, s$,

- for any term t, $[t]$ or $[t]_{\mathcal{A},s}$ denotes t with any function symbols f replaced with $I(f)$ and any variable x replaced with $s(x)$,
- for any atomic formula $p(t_1, t_2, \ldots, t_n)$ for some predicate symbol $p \in P_n$, $\mathcal{A}, s \vDash p(t)$ iff $([t_1], [t_2], \ldots, [t_n]) \in I(p)$, and
- T-schema or truth schema: with $\mathcal{J}$ mapping formula P to 1 if $\mathcal{A}, s \vDash P$ and to 0 if $\mathcal{A}, s \nvDash P$,
  - for any $P \in \mathcal{L}$, $\mathcal{J}(\neg P) = \neg \mathcal{J}(P)$,
  - for any $P, Q \in \mathcal{L}$, $\mathcal{J}(P \wedge Q) = \mathcal{J}(P) \wedge \mathcal{J}(Q)$,
  - for any $P, Q \in \mathcal{L}$, $\mathcal{J}(P \vee Q) = \mathcal{J}(P) \vee \mathcal{J}(Q)$,
  - for any $P, Q \in \mathcal{L}$, $\mathcal{J}(P \implies Q) = \mathcal{J}(P) \implies \mathcal{J}(Q)$,
  - $\mathcal{A}, s \vDash \forall x P$ iff $\forall a \in A$, replacing x with a makes $\mathcal{A}, s \vDash P$, that is, $\forall x P = \bigwedge_{a \in A} P[x \mapsto a]$, and
  - $\mathcal{A}, s \vDash \exists x P$ iff $\exists a \in A$, replacing x with a makes $\mathcal{A}, s \vDash P$, that is, $\exists x P = \bigvee_{a \in A} P[x \mapsto a]$.

For sentences, $\mathcal{A}, s$ can be denoted as $\mathcal{A}$ since there is no free variable and thus s is irrelevant.

iii Semantic consequence

Given formulas $S_1, S_2, \dots S_n, T$, $S_1, S_2, \dots S_n \vDash T$, called that T is a semantic consequence of $S_1, S_2, \dots S_n$, means that for any interpretation $\mathcal{A}, s$, if $\mathcal{A}, s \vDash S_1, \mathcal{A}, s \vDash S_2, \dots, \mathcal{A}, s \vDash S_n$, then $\mathcal{A}, s \vDash T$.

iv Equivalence of syntactic and semantic consequences

For any formulas $S_1, S_2, \dots S_n, T$,

$$S_1, S_2, \dots S_n \vdash T \iff S_1, S_2, \dots S_n \vDash T.$$

Thus syntactic and semantic consequences are called (logical) consequences.

The $\Leftarrow$ direction is the Gödel's completeness theorem and the $\Rightarrow$ direction is from the Henkin's model existence theorem.

v Classification of formulas

A formula P is called (logically) valid or a tautology, denoted as $\top$, iff for any interpretation $\mathcal{A}, s$, $\mathcal{A}, s \vDash P$.

A formula P is called (logically) invalid or a contradiction, denoted as $\bot$, iff for any interpretation $\mathcal{A}, s$, $\mathcal{A}, s \nvDash P$.

A formula P is called contingent iff it is neither a tautology nor a contradiction.

A formula is satisfiable iff it is not a contradiction.

vi Typical rules of inference

- For all equations that hold in Boolean algebra, with each Boolean variable replaced with a metaformula, LHS as numerator and RHS as denominator,
- for any formula P with one free variable, for any variable a,

  $$\frac{\forall x P(x)}{P(a)}$$

  (universal instantiation), and
- for any formula P with one free variable, for any variable a,

  $$\frac{P(a)}{\exists x P(x)}$$

  (existential generalization).

V (Formal) (Deductive) Theory

i (Formal) (deductive) theory

A (formal) (deductive) theory T over a formal language $\mathcal{L}$ is a set of formulas in $\mathcal{L}$ that is closed under $\vdash$. Each $P \in T$ is called a theorem.

ii Deductive system

For a theory T over language $\mathcal{L}$, a deductive system that $T = \{P \in \mathcal{L} \mid\vdash P\}$ is called to derive T and is not distinguished with T in terms of properties.

iii (Syntactical) consistency

A theory T is (syntactically) consistent iff there is no formula P such that $P \in T$ and $\neg P \in T$.

iv Satisfiability

A theory T is satisfiable iff there exists an interpretation $\mathcal{I}$ such that for every $P \in T$, $\mathcal{I} \vDash P$, and T is said to be satisfied by $\mathcal{I}$ or has an interpretation $\mathcal{I}$.

v Complete (consistent) theory

A complete (consistent) theory T is a consistent theory such that for every formula P in its language, either $T \vdash P$ or $T \cup \{P\}$ is inconsistent.

vi Subtheory

If a theory S and a theory T is such that $S \subseteq T$, then S is called a subtheory of T, T is called a supertheory or an extension of S.

vii Effectively axiomatizable, computably axiomatizable, or recursively axiomatizable

A theory T over a language $\mathcal{L}$ is effectively axiomatizable, computably axiomatizable, or recursively axiomatizable if there exists a deductive system with recursively enumerable set of axioms such that $T = \{P \in \mathcal{L} \mid\vdash P\}$, that is, there exists a finitary deductive system such that $T = \{P \in \mathcal{L} \mid\vdash P\}$ and there exists a recursively enumerable set of formal proofs that derive all $P \in T$.

viii Independent formula

For any independent formula P of a sound deductive system that generate a satisfiable theory T, $T \cup \{P\}$ and $T \cup \{\neg P\}$ are both satisfiable.

ix (Complete) theory of interpretation

The (complete) theory of an interpretation $\mathcal{A}$ is all formulas that are satisfied by $\mathcal{A}$, denoted as $\text{Th}(\mathcal{A})$.

The theory of a class K of interpretations is all formulas that are satisfied by all $k \in K$, denoted as $\text{Th}(K)$.

VI First-order theory

i First-order theory

A (formal) (deductive) first-order theory T over a formal language $\mathcal{L}$ is a set of sentences in $\mathcal{L}$ that is closed under $\vdash$.

ii First-order theory with identity

A first-order theory is with identiy if its language has the equality symbol $=$, and the set of theorems of the theory contains all applications of the following axiom schemas:

- Reflexivity: For any variable x, $x = x$.
- Leibniz's law or substitution: For any $n \in \mathbb{N}_0$, for any formula with any n free variables $x_1, x_2, \dots, x_n$, any $i \in \mathbb{N} \wedge i \leq n$, and any variable x,

$$(x_i = x) \implies (P(x_1, x_2, \dots, x_i, \dots, x_n) \implies P(x_1, x_2, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n)).$$

iii Equivalence of consistency and satisfiability

A theory over proportional or first-order language is consistent iff it is satisfiable.

The $\Leftarrow$ direction is from the Gödel's completeness theorem and the $\Rightarrow$ direction is the Henkin's model existence theorem.

iv Elementary class or axiomatizable class

The elementary class or axiomatizable class of a first-order theory T is the class of all structures that satisfies T.

v Downward Löwenheim–Skolem Theorem

If a first-order theory with signature σ has an infinite model $\mathcal{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every μ such that $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$, it has model of cardinality μ that is an elementary substructures of $\mathcal{M}$.

vi Upward Löwenheim–Skolem Theorem

If a first-order theory with signature σ has an infinite model $\mathcal{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every $\mu \geq \kappa$, it has model of cardinality μ that is an elementary extension of $\mathcal{M}$.

vii Compactness theorem

If a first-order theory T is such that all its finite subsets have models, then T has model.

VII Gödel's incompleteness theorems

i Gödel's first incompleteness theorem

Any recursively enumerable consistent deductive system that can interprets the Robinson arithmetic is syntactically incomplete.

ii Gödel's second incompleteness theorem

Any recursively enumerable consistent deductive system that can interprets the Robinson arithmetic can not prove its consistency.

VIII Abstract logic

i Abstract logic

Let L be a formal language closed under logical connectives and with functionally complete set of logical connectives, and $\mathscr{M}$ be all models (possibly designated to be allowed) for it. The ordered pair $(L, \mathscr{M})$ is called an abstract logic.

ii Isomorphism-invariant

An abstract logic, $(L, \mathscr{M})$, is isomorphism-invariant iff for every two models $\mathcal{M}, \mathcal{N} \in \mathscr{M}$, for every sentence $\varphi \in L$, φ is isomorphism-invariant between $\mathcal{M}$ and $\mathcal{N}$.

iii Expressive power

All isomorphism-invariant abstract logics form a class $\mathbb{L}$ with partial order $\leq$, called weaker than or less expressive than, equivalence relation $=$, and strict partial order $<$, called strictly weaker than or strictly less expressive than, of expressive power defined, for any $(L_1, \mathscr{M}_1), (L_2, \mathscr{M}_2) \in \mathbb{L}$, with the $\mathscr{N}_1$ and $\mathscr{N}_2$ defined such that all elements of $\mathscr{N}_1 \cup \mathscr{N}_2$ have the same signature σ and domain N, that for any $\mathcal{M} \in \mathscr{M}_1$, there exists at least one $\mathcal{N} \in \mathscr{N}_1$ such that there exists at least one isomorphism between $\mathcal{M}$ and $\mathcal{N}$, and that for any $\mathcal{M} \in \mathscr{M}_2$, there exists at least one $\mathcal{N} \in \mathscr{N}_2$ such that there exists at least

one isomorphism between $\mathcal{M}$ and $\mathcal{N}$, as:

$$(L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \iff \bigcup_{\mathcal{N} \in \mathscr{N}_1} \text{Def}(\mathcal{N}) \subseteq \bigcup_{\mathcal{N} \in \mathscr{N}_2} \text{Def}(\mathcal{N}),$$

$$(L_1, \mathscr{M}_1) = (L_2, \mathscr{M}_2) \iff (L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \wedge (L_2, \mathscr{M}_2) \leq (L_1, \mathscr{M}_1).$$

$$(L_1, \mathscr{M}_1) < (L_2, \mathscr{M}_2) \iff (L_1, \mathscr{M}_1) \leq (L_2, \mathscr{M}_2) \wedge \neg(L_2, \mathscr{M}_2) \leq (L_1, \mathscr{M}_1).$$

iv Compactness property

For an abstract logic $(L, \mathscr{M})$, if for every theory T over its language, that all its finite subsets have model in $\mathscr{M}$ implies that T has model in $\mathscr{M}$, then the abstract logic has compactness property.

Compactness property implies that for any theory has model in $\mathscr{M}$, there exists a sound finitary deductive system that derive it.

That for any theory of a model in $\mathscr{M}$, there exists a sound, finitary and semantically complete deductive system that derive it, implies compactness property.

v Downward Löwenheim–Skolem property

For an abstract logic $(L, \mathscr{M})$ with signature σ, if for every theory T over L that has an infinite model $\mathcal{M} \in \mathscr{M}$ of some cardinality $\kappa \geq \aleph_0$, for every μ such that $\max(|\sigma|, \aleph_0) \leq \mu \leq \kappa$, it has model of cardinality μ that is an elementary substructures of $\mathcal{M} \in \mathscr{M}$, then the abstract logic is called to have downward Löwenheim–Skolem property.

vi Upward Löwenheim–Skolem Theorem

For an abstract logic $(L, \mathscr{M})$, if for every theory T over L that has an infinite model $\mathcal{M} \in \mathscr{M}$ of some cardinality $\kappa \geq \aleph_0$, then for every $\mu \geq \kappa$, it has model of cardinality μ that is an elementary extension of $\mathcal{M} \in \mathscr{M}$.

vii Lindström's theorem

Any isomorphism-invariant abstract logic extending first-order logic with standard semantics that has compact property and downward Löwenheim–Skolem property must not has strictly more expressive power than first-order logic with standard semantics.